Practice Quiz: Thinking Like an Inventor

Part A: Multiple Choice

1. In Active Inference, cognition is best described as: A) Random generation of ideas followed by selection B) Inference over hidden causes — reasoning backward from observed symptoms to underlying causal structure C) Pure logical deduction from first principles D) Unconscious pattern matching without explicit reasoning

2. Bela Barenyi's invention of crumple zones required which cognitive move? A) Making the car chassis stiffer and heavier B) Reframing the generative model from "rigidity protects" to "controlled deformation absorbs energy" C) Copying an existing design from aircraft engineering D) Increasing engine power to avoid collisions

3. Functional fixedness in invention refers to: A) The inability to repair a broken mechanism B) The inability to see an object or process being used for purposes other than its conventional one C) A design that is permanently fixed in place D) A constraint imposed by building codes

4. An inventor mentally simulating how their device would work is using: A) Random trial and error B) Counterfactual reasoning through their generative model to predict outcomes of design changes C) Passive observation of existing devices D) Pure mathematical calculation without visualization

5. The Five Whys technique helps inventors: A) Ask five different people what they think the problem is B) Trace symptoms backward through causal chains to identify root causes C) Generate five different product names D) Test five different prototypes

6. Bayesian model comparison in inventive cognition involves: A) Choosing the first explanation that comes to mind B) Evaluating evidence for and against multiple competing causal models before committing to one C) Comparing your invention to five competing products D) Using only statistical analysis, never intuition

7. When an inventor moves from asking "how do I build a better mousetrap?" to "how do I keep mice out of the pantry?", they are applying: A) The constraint addition technique B) Functional fixedness C) Problem abstraction to a higher level, opening new solution spaces D) Einstellung — applying a familiar method to a new problem

Part B: Short Answer and Analysis

1. A product designer is trying to make a more comfortable bicycle seat. After months of iterating on padding materials and seat shapes with limited success, they are stuck. Describe how each of the three reframing techniques (inversion, constraint removal, abstraction) could help them escape this fixation. For each technique, state the reframed question and suggest one new solution direction it opens.

2. Nikola Tesla famously ran mental simulations of his inventions in extraordinary detail, while Thomas Edison famously favored physical trial and error. Using the Active Inference concept of generative model accuracy, explain (a) the conditions under which Tesla's approach is superior, (b) the conditions under which Edison's approach is superior, and (c) what an ideal inventive cognitive process would combine from both approaches.

3. An inventor observes that their new solar-powered garden light stops working after three months. Three possible hidden causes are: (a) the battery degrades from heat cycling, (b) the solar panel accumulates dirt that blocks light, and (c) the circuit board corrodes from moisture. Design a diagnostic process using Bayesian model comparison. For each candidate cause, state what specific observation would increase or decrease the probability of that cause, and describe the single most decisive test that would distinguish between the three.